

Do Artificial Stakes Improve LLM Risk Reasoning?

CS 639: Introduction to Foundation Models

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1. Overarching Idea and Motivation

LLMs are being used more for reasoning in high-stakes domains like finance and risk assessment, but they never face real consequences for bad choices. They’re overconfident across model families and sizes (Cheng et al. 2025), and LLM financial agents can’t even beat a simple buy-and-hold strategy (Li et al., ACL 2025). RL on reasoning steps runs into reward hacking and local maxima where models overoptimize individual steps rather than the full reasoning chain (Weng 2024), and models can generate convincing CoT traces that don’t reflect their internal processes (Lanham et al., Anthropic 2023).

We want to test a simpler alternative: give the model a persistent virtual budget that depletes based on decision quality across a sequence of problems. Budget-aware token control already changes how LLMs reason (BudgetThinker, 2024), and LLMs exhibit behavioral economics biases like loss aversion (Ross et al. 2024), but testing whether persistent resource stakes actually improve decision quality on tasks with verifiable answers hasn’t yet been done.

2. Hypothesis

We believe that persistent resource framing—where the model sees a visible budget that updates based on its decision quality across a sequence of problems—will lead to better risk reasoning compared to matched sequential baselines receiving the same feedback but no stakes. We expect to see higher decision accuracy against ground truth and better probability calibration. Importantly, we want to make sure any gains reflect the model actually deciding differently, not just describing its reasoning differently. We also want this to hold across multiple independent runs under the same consistent stakes and system prompt.

3. Methodology

Datasets. We use three dataset types. (1) *Synthetic*: expected value and probability problems with derivable correct answers, providing clean ground truth in a controlled setting. (2) *Real-world*: ConvFinQA (financial reasoning), ARC-Challenge (logical reasoning), and MedQA (medical decision-making), to test whether effects generalize beyond synthetic tasks. (3) *Sequential*: a small multi-step budget allocation environment where outcomes are sampled from a known distribution and the budget updates over time, purpose-built to test persistence and whether early mistakes compound.

Experimental Design. Each dataset is tested under four conditions: bare baseline (no framing), surface stakes (one-shot budget mention), persistent stakes (budget carries across decisions and updates based on outcomes), and CoT-only control (no stakes language). We collect multiple independent samples per condition and compare single-sample accuracy, majority vote, and best-of- n by confidence.

Metrics. Decision quality via accuracy on multiple-choice tasks and regret for sequential tasks. Calibration via calibration curves, ECE, and Brier score. Reasoning proxies include number of alternatives considered, risk factors mentioned, and logical error rate from CoT traces. For sequential tasks: cumulative reward/regret, budget efficiency, recovery rate, and mistake propagation via $P(\text{mistake}_{t+1} \mid \text{mistake}_t)$ minus baseline. Claude Sonnet serves as LLM-as-a-Judge for coherence and reasoning coverage.

Models. Llama-3.1-8B Base vs. Instruct via Ollama/vLLM, to study how post-training affects sensitivity to stakes framing. Claude Sonnet as automated evaluator.

Tools and Infrastructure. Python, pandas/numpy/scipy/matplotlib, Anthropic SDK, Ollama, HuggingFace Transformers, vLLM. Local inference on an NVIDIA RTX 6000 GPU. Sequential environments, budget tracking, and API calls managed through standard Python tooling.

4. Plan of Action

- Finalize experimental scope, tasks, conditions, and evaluation approach
- Build dataset pipeline (synthetic + real-world + sequential environment)
- Run initial experiments and adjust methodology based on feasibility
- Full experimental run across conditions and models
- Analyze results, write up findings, and submit report by May 7

[1] Cheng et al., “Mind the Confidence Gap,” arXiv:2502.11028, 2025. [2] Li et al., “InvestorBench,” ACL 2025. [3] Weng, “Reward Hacking in RL,” *Lil’Log*, 2024. [4] Lanham et al., “Measuring Faithfulness in CoT,” Anthropic, 2023. [5] “BudgetThinker,” arXiv:2508.17196, 2024. [6] Ross et al., “LLM Economicus?,” arXiv:2408.02784, 2024.